CMI Entrance Examination

Mock Test · Part B

Question 1

15 points

Let $f : \mathbb{R} \to \mathbb{R}$ be a function. We say f has a *discontinuity* at $x = a$ if it is not continuous there. Discontinuities are broadly classified into three types:

- **Removable Discontinuity:** $\lim_{x \to a} f(x)$ exists, but it is not equal to $f(a)$ (or $f(a)$ is undefined).
- **Jump Discontinuity:** Both one-sided limits $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ exist and are finite, but they are not equal.
- **Essential Discontinuity:** At least one of the one-sided limits $\lim_{x \to a^-} f(x)$ or $\lim_{x \to a^+} f(x)$ does not exist (or is infinite).

(a) Consider the function $f(x)$ defined as $f(x) = x \sin(1/x)$ for $x \neq 0$, and $f(0) = 1$. Classify the type of discontinuity this function has at $x = 0$. Justify your answer using limits.

(b) A function f is said to satisfy the *Intermediate Value Property* (IVP) on an interval I if, for any two points $a, b \in I$ and any real number k strictly between $f(a)$ and $f(b)$, there exists at least one point c between a and b such that $f(c) = k$. Determine whether the function $f(x)$ defined in part (a) satisfies the IVP on the interval $[-1, 1]$. Justify your answer.

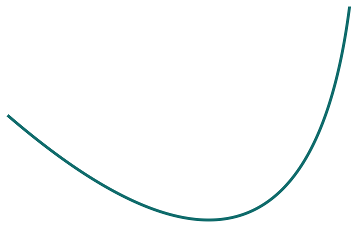
(c) Prove that if a function $g : \mathbb{R} \to \mathbb{R}$ has a discontinuity at $x = a$ but satisfies the Intermediate Value Property on any interval containing a, then the discontinuity at $x = a$ must be an essential discontinuity.

Question 2

15 points

A *straightedge* is a basic geometric tool—like an unmarked ruler—that can only be used to draw a straight line passing through two given points. It cannot be used to measure distances.

You are given an arbitrarily oriented parabola drawn on a standard Euclidean plane, as illustrated below.



Using only a straightedge, a compass, and a pencil, provide a step-by-step geometric construction to draw the axes of the given parabola.

Question 3

15 points

The Binomial Theorem states that for any real numbers x, y and any positive integer n, the expansion is given by:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

(a) Does this theorem universally hold true for any $n \times n$ square matrices A and B with real entries? That is, is it always true that

$$(A+B)^n = \sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k\,?$$

If yes, provide a brief justification. If no, provide an explicit counterexample using 2×2 matrices for $n = 2$.

(b) Now suppose we are given the additional condition that matrices A and B commute (i.e., $AB = BA$). Prove that under this condition, the binomial expansion will correctly evaluate $(A+B)^n$ for all positive integers n.

Question 4

15 points

Let A_n denote the number of ways to tile a $2 \times n$ rectangular grid using any combination of 1×2 dominoes (which can be placed either vertically or horizontally) and 2×2 square tiles.

(a) Compute A_1, A_2, and A_3 by explicitly counting the configurations.

(b) Establish a linear recurrence relation for A_n in terms of A_{n-1} and A_{n-2} for $n \geq 3$. Provide a combinatorial argument justifying your recurrence.

(c) Using your recurrence relation, prove by mathematical induction (or by generating functions) that:

$$A_n = \frac{2^{n+1} + (-1)^n}{3}$$

for all $n \geq 1$.

Question 5 *15 points*

Let $f : [0,1] \to \mathbb{R}$ be a continuously differentiable function such that $f(0) = 0$ and the integral of its squared derivative strictly satisfies:

$$\int_0^1 \left(f'(x)\right)^2 dx \;=\; 1$$

(a) Using the Cauchy–Schwarz inequality for integrals (or otherwise), prove that for all $x \in [0,1]$, the function is bounded by $|f(x)| \le \sqrt{x}$.

(b) Using the result from part (a), deduce that:

$$\int_0^1 f(x)\,dx \;\le\; \frac{2}{3}$$

(c) Determine whether equality can hold in the bound established in part (b). If equality can hold, find the exact function $f(x)$. If it cannot hold, precisely explain why no such function f exists.

Question 6 *15 points*

Let $p > 3$ be a prime number.

(a) Evaluate the sum $1^2 + 2^2 + \cdots + (p-1)^2$ and deduce that it is divisible by p.

(b) Consider the polynomial $P(x) = (x-1)(x-2)\cdots(x-(p-1))$. Let S_1 be the sum of its roots, and S_2 be the sum of the products of its roots taken two at a time. Using part (a) or otherwise, prove that S_1 and S_2 are both divisible by p.

(c) Let the harmonic number

$$H_{p-1} \;=\; 1 + \tfrac{1}{2} + \tfrac{1}{3} + \cdots + \tfrac{1}{p-1}$$

be expressed as a single fraction $\dfrac{A}{B}$ in lowest terms. Using the properties of the polynomial $P(x)$ modulo p, prove that the numerator A is divisible by p. Is A divisible also by p^2? Justify.

End of Part B